

Chris Rockwell

3150 S. Delaware Street
San Mateo, CA 94403
chris.rockwell8@gmail.com
<https://crockwell.github.io/>

PROFESSIONAL EXPERIENCE	Roblox <i>AI Scientist, Foundational AI</i> Jun. 2026 - Present San Mateo, CA • Multimodal generative AI towards photorealistic gaming Morpheus AI <i>Member of Technical Staff</i> Sep. 2025 - May 2026 Pittsburgh, PA • Led data and evaluation, contributed to modeling and presentation for two successful real-time video generation and world model projects (company + IP acquired by Roblox) NVIDIA <i>Research Intern, Deep Imagination Research Group</i> Mar. 2024 - Oct. 2024 Santa Clara, CA • Systematically curated and annotated cameras on 100K diverse, dynamic Internet videos Meta <i>Research Scientist Intern, Computational Photography Research</i> May 2022 - Dec. 2022 Seattle, WA • Produced lightweight radiance field conditioned upon a single image TuringSense <i>Technical Consultant, Computer Vision</i> Feb. 2021 - Apr. 2021 Santa Clara, CA • Suggested and implemented improvements to TuringSense home yoga product Citadel <i>Trader, Global Fixed Income</i> Apr. 2017 - Oct. 2017 New York, NY • Designed and executed quantitative trading strategies to enhance team's portfolio BNP Paribas <i>Analyst, Fixed Income Structuring</i> Jul. 2015 - Mar. 2017 New York, NY • Created systematic hedging strategies and priced bespoke options for institutional clients
EDUCATION	University of Michigan <i>Ph.D. in Computer Science and Engineering</i> Sep. 2020 - Aug. 2025 Ann Arbor, MI • Advisors: Justin Johnson, David F. Fouhey <i>Master of Science, Computer Science and Engineering</i> Sep. 2018 - May 2020 • GPA: 4.00/4.00 • Advisors: David F. Fouhey, Jia Deng <i>Bachelor of Science, Economics</i> Sep. 2011 - May 2015 <i>Minors in Computer Science, Mathematics</i> • GPA: 3.95/4.00
INTERESTS	Computer Vision, Machine Learning

PUBLICATIONS	Dynamic Camera Poses and Where to Find Them Chris Rockwell, Joseph Tung, Tsung-Yi Lin, Ming-Yu Liu, David F. Fouhey and Chen-Hsuan Lin CVPR, 2025 Project Page
--------------	--

FAR: Flexible, Accurate and Robust 6DoF Relative Camera Pose Estimation
Chris Rockwell, Nilesh Kulkarni, Linyi Jin, JJ Park, Justin Johnson and David F. Fouhey
CVPR, 2024 (**Highlight**)

[Project Page](#)

Scalable 3D Captioning with Pretrained Models
Tianghe Luo*, **Chris Rockwell***, Honglak Lee[†] and Justin Johnson[†]
NeurIPS (Datasets and Benchmarks Track) 2023

[Project Page](#)

The 8-Point Algorithm as an Inductive Bias for Relative Pose Prediction by ViTs
Chris Rockwell, Justin Johnson and David F. Fouhey
3DV 2022

[Project Page](#)

PlaneFormers: From Sparse View Planes to 3D Reconstruction
Samir Agarwala, Linyi Jin, **Chris Rockwell** and David F. Fouhey
ECCV 2022

[Project Page](#)

FWD: Real-time Novel View Synthesis with Forward Warping and Depth
Ang Cao, **Chris Rockwell** and Justin Johnson
CVPR 2022

[Project Page](#)

Understanding 3D Object Articulation in Internet Videos
Shengyi Qian, Linyi Jin, **Chris Rockwell**, Siyi Chen and David F. Fouhey
CVPR 2022

[Project Page](#)

PixelSynth: Generating a 3D-Consistent Experience from a Single Image
Chris Rockwell, David F. Fouhey and Justin Johnson
ICCV 2021

[Project Page](#)

Full-Body Awareness from Partial Observations
Chris Rockwell and David F. Fouhey
ECCV 2020

[Project Page](#)

RESEARCH
EXPERIENCE

Morpheus AI

Pittsburgh, PA

Member of Technical Staff | Founder: Xun Huang

Sep. 2025 - May 2026

Real-Time Video Generation

- Led data curation and evaluation for real-time video and world models

NVIDIA, Deep Imagination Research Group

Santa Clara, CA

Research Intern | Hosts: Chen-Hsuan Lin, Tsung-Yi Lin

Mar. 2024 - Oct. 2024

Internet Scale 3D Curation – *Dynamic Camera Poses*

- Curate and annotate cameras for 100K dynamic Internet videos from 3.2M diverse videos

Meta Reality Labs, Computational Photography Research

Seattle, WA

Research Scientist Intern | Hosts: Hung-Yu Tseng, Jia-Bin Huang

May 2022 - Dec. 2022

Novel View Synthesis

- Produce lightweight radiance field conditioned upon a single image

Michigan Vision Lab*Graduate Research Assistant* | Advisor: Justin JohnsonAnn Arbor, MI
May 2020 - Aug. 2025Internet Scale 3D-Text Modeling – *Scalable 3D Captioning*

- Apply powerful VLM pipeline to caption 660K 3D assets, finetune text-to-3D models

Novel View Synthesis

- *PixelSynth*: Introduce powerful generative model, enabling 3D-consistent extrapolation
- *FWD*: Real-time NVS using pointcloud and transformer; predecessor to 3DGS

Fouhey AI Lab*Graduate Research Assistant* | Advisor: David F. FouheyAnn Arbor, MI
May 2019 - Aug. 2025

Relative Camera Pose Estimation

- *8-Point ViT*: Include 8-Point machinery in ViT block to improve relative pose estimation
- *FAR*: Fuse correspondence and learning-based pipeline, yielding best-of-both estimates

3D Reconstruction – *PlaneFormers*

- Introduce transformer to learn to refine planar reconstruction

3D Object Articulation – *Understanding 3D Object Articulation*

- Collect rich dataset of people articulating objects and learn axes of object articulation

3D Human Pose Estimation – *Full-Body Awareness*

- Propose self-training method to substantially improve human pose on internet video

Princeton Vision and Learning Lab*Graduate Research Assistant* | Advisor: Jia DengPrinceton, NJ
May 2018 - May 2019

2D Human Pose Estimation

- Add bottleneck-to-attention module to improve *Stacked Hourglass* accuracy 0.7%

Meta-Learning

- Improve finetune model to within 0.1 *avg. rank* of meta-learning baseline on *Meta-Dataset*

Strategic Reasoning Group*Undergraduate Research Assistant* | Advisor: Michael P. WellmanAnn Arbor, MI
May 2013 - Jul. 2013

Agent-based simulation of High-Frequency Trading and Latency Arbitrage

- Model trading agents with varying speeds to measure effects of latency arbitrage

SERVICE

Reviewer: CVPR (2023 Outstanding Reviewer), NeurIPS (2023 Top Reviewer), ICCV, ECCV, 3DV, ICLR, ICML, TPAMI

AI4ALL Project Instructor: lead vision project for nine underrepresented high-schoolers

AI4ALL Curriculum Advisory Board Member: contributed to national curriculum

Technical Mentor: mentored five students with David F. Fouhey, including one in African Undergraduate Research Adventure (AURA)

Graduate Student Advisory Committee: represented CSE students to improve experience